

Experimental Design - Practice Multiple Choice Questions

Exam 1 Preparation: CRD, RCBD, and Factorial Designs

AGST 50104 - Spring 2026

2026-02-01

Instructions

- Each question has **4 options**
 - Only **one option is correct**
 - Read experiment descriptions carefully for degrees of freedom calculations
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Question 1

A researcher tests 5 nitrogen fertilizer rates on wheat yield using 8 replicate plots per rate in a completely randomized design (CRD). What are the degrees of freedom for **error** in the ANOVA table?

- a) 36
 - b) 7
 - c) 35
 - d) 39
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Question 2

An agronomist conducts a field trial with 4 treatments arranged in 6 complete blocks. Each block contains all 4 treatments randomized within the block. What are the degrees of freedom for **treatments** in the RCBD ANOVA?

- a) 3
 - b) 5
 - c) 15
 - d) 23
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Question 3

A greenhouse study tests the effect of 3 irrigation levels on tomato plants. The researcher places 5 plants in each of 4 pots per irrigation level. At harvest, all 20 tomatoes per pot are weighed. What is the correct **experimental unit** in this study?

- a) Individual tomato fruit
 - b) Individual tomato plant
 - c) Pot
 - d) Irrigation level
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Question 4

In the greenhouse study described in Question 3, the researcher reports “ $n = 20$ tomatoes per treatment” and runs an ANOVA treating each tomato as an independent replicate. What is the **problem** with this analysis?

- a) Sample size is too small
 - b) ANOVA cannot be used for plant data
 - c) Pseudo-replication: multiple tomatoes per pot are not independent
 - d) Irrigation levels should be blocked
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Question 5

A livestock nutritionist tests 4 diets on cattle. She has 24 animals and randomly assigns 6 animals to each diet, with each animal housed in a separate pen. Which experimental design is this?

- a) Factorial design
- b) Completely Randomized Design (CRD)
- c) Randomized Complete Block Design (RCBD)
- d) Split-plot design

Question 6

An orchard experiment compares 5 apple tree varieties. The orchard has a known fertility gradient from north to south. The researcher divides the orchard into 5 strips (north to south) and plants one tree of each variety randomly within each strip. What design is this?

- a) Completely Randomized Design (CRD)
 - b) Randomized Complete Block Design (RCBD)
 - c) Latin Square Design
 - d) Factorial design
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Question 7

In an RCBD with 4 blocks and 5 treatments, what are the degrees of freedom for **error**?

- a) 4
 - b) 12
 - c) 16
 - d) 19
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Question 8

A factorial experiment tests 3 levels of nitrogen (0, 50, 100 kg/ha) and 2 levels of phosphorus (0, 30 kg/ha) with 4 replicates per combination in a CRD structure. What are the degrees of freedom for **error** in the ANOVA?

- a) 5
- b) 12
- c) 18

d) 23

Question 9

A researcher runs a Q-Q plot of residuals from an ANOVA and observes that points form an S-curve (systematic deviation from the diagonal line). What does this indicate?

- a) Homogeneity of variance is violated
 - b) Residuals are skewed (non-normal)
 - c) Independence assumption is violated
 - d) The model fits perfectly
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Question 10

Which diagnostic plot is **most useful** for detecting heteroscedasticity (unequal variances)?

- a) Q-Q plot
 - b) Residuals vs. Fitted Values plot
 - c) Histogram of residuals
 - d) Cook's distance plot
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Question 11

An ANOVA F-test for treatments yields $p = 0.001$ (highly significant). Which statement is **correct**?

- a) All treatment means are significantly different from each other
- b) The largest mean is significantly different from the smallest mean

- c) At least one treatment mean differs from the others
 - d) The treatment effect is practically important
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Question 12

Which post-hoc test is **most appropriate** when you want to compare all treatment means to a single **control group** only (not all pairwise comparisons)?

- a) Tukey HSD
 - b) Bonferroni
 - c) Fisher's LSD
 - d) Dunnett's test
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Question 13

A 2×3 factorial experiment has 2 levels of tillage (conventional, no-till) and 3 levels of cover crop (none, rye, vetch). The ANOVA shows a significant **Tillage** $\times$ **Cover Crop** interaction ($p = 0.003$). How should you interpret the results?

- a) Report only the main effect of tillage
 - b) Report only the main effect of cover crop
 - c) Report both main effects; ignore the interaction
 - d) Examine simple effects; interaction takes precedence over main effects
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Question 14

An ANOVA F-test yields $F = 9.0$ with $df = (5, 24)$ and $p = 0.00012$. What is the **correct interpretation** of this p-value?

- a) There is a 0.012% chance the null hypothesis is true
 - b) There is a 99.988% chance the treatment effects are real
 - c) If H_0 were true, there's a 0.012% chance of observing an F-statistic this large or larger
 - d) The probability that all treatment means are equal is 0.00012
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Question 15

A field trial with 4 treatments and 6 complete blocks (RCBD) shows $SS_{Treatment} = 240$, $SS_{Block} = 180$, $SS_{Error} = 120$. What is MS_{Error} ?

- a) 6
 - b) 8
 - c) 10
 - d) 15
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Question 16

When should you use **blocking** in an experimental design?

- a) When you want to increase the number of treatments tested
 - b) When there is a known source of variation you want to control
 - c) When sample size is too small for a CRD
 - d) When you need to test interactions between factors
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Question 17

What is the **primary benefit** of randomization in experimental design?

- a) It increases sample size
 - b) It ensures equal group sizes
 - c) It minimizes bias by ensuring treatment assignment is not confounded with unknown factors
 - d) It guarantees statistically significant results
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Question 18

A food scientist wants to study the effects of baking temperature and mixing time on cake texture. She considers two approaches: (1) **One-factor-at-a-time (OFAT)**: test 3 temperatures at fixed mixing time, then test 3 mixing times at fixed temperature; or (2) **Factorial design**: test all 9 combinations of 3 temperatures $\times$ 3 mixing times. What is the **main advantage** of the factorial design over OFAT?

- a) Factorial designs require more experimental units
 - b) Factorial designs are easier to analyze statistically
 - c) Factorial designs can detect interactions between factors
 - d) Factorial designs always yield significant results
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Question 19

Fisher's three principles of experimentation are:

- a) Sampling, estimation, hypothesis testing
 - b) Normality, independence, homogeneity
 - c) Replication, randomization, blocking
 - d) Power, precision, bias
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Question 20

A bakery researcher wants to optimize bread quality by testing 3 flour types and 3 yeast strains. Using a **one-factor-at-a-time (OFAT)** approach with 4 replicates per treatment, she would test 3 flour types (holding yeast constant) then 3 yeast strains (holding flour constant) = 24 total loaves. If she instead uses a **3×3 factorial design** with 4 replicates per combination, how many loaves would she bake, and what is the key efficiency advantage?

- a) 36 loaves; each main effect is estimated from all 36 observations (hidden replication), not just 12
 - b) 24 loaves; factorial designs use the same number of runs as OFAT
 - c) 36 loaves; factorial designs always detect interactions
 - d) 9 loaves; factorial designs are more efficient than OFAT
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Bonus Question 21

An RCBD experiment with 5 treatments and 4 blocks yields the following ANOVA table:

Source	df	SS	MS	F
Block	3	120	40	-
Treatment	4	480	120	10.0
Error	12	144	12	-
Total	19	744	-	-

If we had analyzed this same data as a **CRD** (ignoring blocks), what would happen to the **error term**?

- a) Error would decrease
- b) Error would stay the same
- c) Error would increase (SS_Error would include SS_Block)
- d) Cannot determine from given information